

Rajiv Gandhi University of Health Sciences, Karnataka
Fourth Semester B. Pharm Degree Examination – 16-Nov-2022

Time: Three Hours

Max. Marks: 75 Marks

PHARMACEUTICAL ORGANIC CHEMISTRY - III

Q.P. CODE: 5013

Your answers should be specific to the questions asked.

Draw neat labeled diagrams wherever necessary.

All the Questions are compulsory.

LONG ESSAYS

2 x 10 = 20 Marks

1. Define configuration. Discuss the D-L and R-S system of nomenclature in detail.
OR
Define conformations. Write about the factors affecting stability of conformations. Explain the conformations in cyclohexane and their energy relationships.
2. Describe the various methods of preparations and chemical reactions of thiophene. Give the structure and medicinal uses of any two thiophene analogs.

SHORT ESSAYS

7 x 5 = 35 Marks

3. Explain the different types of elements of symmetry with suitable examples.
OR
Discuss any three reactions of chiral molecules.
4. What are geometrical isomers? Discuss their conditions and nomenclature.
OR
What are heterocyclic compounds? Discuss the nomenclature of them.
5. Discuss the aromatic behaviour of thiophene. Explain how its aromaticity is compared with Furan.
6. Write any two methods of synthesis and chemical reactions of thiazole.
7. Write any two methods of synthesis and chemical reactions of quinoline.
8. What is Beckmann's rearrangement? Give its applications and mechanism.
9. Define and explain Clemmenson's reduction with examples and mechanism.

SHORT ANSWERS

10 x 2 = 20 Marks

10. Define the terms a) Optical activity b) Chirality.
11. Differentiate plane polarized light and monochromatic light.
12. Define Atropisomerism. Give its conditions.
13. What are Stereo selective reactions? Give an example.
14. Give reason why pyrrole behaves as a weak base.
15. Write the structure and medicinal use of any two furan derivatives.
16. Write the structure and medicinal use of any two imidazoles.
17. Write the reactions in Fisher indole synthesis.
18. What is Birch reduction? Give an example.
19. Write Dakin's reduction. Give an example.
